

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B6: CONDENSED-MATTER PHYSICS

TRINITY TERM 2018

Friday, 8 June, 2.30 pm – 4.30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. (a) Define the term *lattice* as used to describe crystal structure.

The element tungsten in its usual crystalline form (known as α -tungsten) has a cubic conventional unit cell with tungsten atoms at positions $[0, 0, 0]$ and $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ in units of the lattice constant. What is the lattice type of α -tungsten?

[4]

(b) X-rays of wavelength 0.1542 nm are incident on a powdered sample of α -tungsten, and the first X-ray diffraction ring (lowest angle of deflection) is observed at a deflection angle of 40.26° . Determine the lattice constant and the density of α -tungsten given that its atomic weight is 183.8 g mol^{-1} .

[9]

(c) Occasionally, elemental tungsten can be found in a different crystal form, known as β -tungsten. This has a cubic conventional unit cell with tungsten atoms at positions

$$[0, 0, 0], \quad [\frac{1}{2}, \frac{1}{2}, \frac{1}{2}], \quad [\frac{1}{4}, \frac{1}{2}, 0], \quad [\frac{3}{4}, \frac{1}{2}, 0], \quad [\frac{1}{2}, 0, \frac{1}{4}], \quad [\frac{1}{2}, 0, \frac{3}{4}], \quad [0, \frac{1}{4}, \frac{1}{2}], \quad [0, \frac{3}{4}, \frac{1}{2}].$$

What is the lattice type of this crystal?

[2]

(d) In an X-ray scattering experiment with a powdered β -tungsten sample, what are the Miller indices associated with the first two (lowest angle) non-vanishing diffraction rings? Estimate the ratio of measured X-ray intensities in these first two non-vanishing diffraction rings.

[7]

(e) With very careful experimentation in very pure β -tungsten, it is expected that one or more additional diffraction rings at even lower scattering angles will appear with intensity much lower than the easily observed peaks we have discussed above. Giving your reasoning, what are the Miller indices of an additional observable diffraction ring, and why might it be seen?

[3]

2. (a) Define the terms *chemical potential* and *Fermi energy*. [2]

(b) Consider a one-dimensional tight binding model of electrons hopping on a one-dimensional crystal of identical atoms with lattice constant a and one atom per unit cell. Write the wavefunction of the atomic orbital on atom n as $|n\rangle$ and assume $\langle n|m\rangle = \delta_{nm}$. The matrix elements of the Hamiltonian H are given by

$$\begin{aligned}\langle n|H|n\rangle &= \epsilon_0 \\ \langle n|H|n+1\rangle &= \langle n+1|H|n\rangle = -t \\ \langle m|H|n\rangle &= 0 \quad \text{for } |m-n| > 1.\end{aligned}$$

By writing an appropriate Schrödinger equation, show that the dispersion relation for the electrons in this system is given by

$$E(k) = \epsilon_0 - 2t \cos(ka)$$

where k is the wavevector. Make a labeled diagram of this dispersion relation in the first Brillouin zone. Make a labeled sketch of the density of states. [9]

(c) Assume each atom is monovalent. Determine the Fermi energy and, without detailed calculation, show that the chemical potential is independent of temperature. [5]

$$\left[\text{Hint: you may find it helpful to use the identity } \frac{1}{e^{\beta(E-\mu)} + 1} = 1 - \frac{1}{e^{\beta(\mu-E)} + 1} \right]$$

(d) Consider now a two-dimensional crystal having a square lattice with lattice constant a and a single atom per unit cell. Suppose the energy as a function of wavevector is given by

$$E(k_x, k_y) = \epsilon_0 - 2t[\cos(k_x a) + \cos(k_y a)]$$

with $t > 0$. Again assuming that the atoms are monovalent, draw and label the Brillouin zone and the Fermi surface. What is the chemical potential as a function of temperature? [4]

(e) Suppose that the same two-dimensional crystal is squeezed in one direction so that it now has a rectangular lattice, with lattice constant a_x in the x -direction, and a_y in the y -direction with $a_x < a_y$. Draw the Brillouin zone and make a sketch of a Fermi surface that would be consistent with the information given. [5]

3. (a) Define the terms *phonon* and *Brillouin zone*. [2]

(b) The energy V of two lead (Pb) atoms as a function of their separation is often approximated by the Morse potential:

$$V(r) = Ae^{-2r/\xi} - Be^{-r/\xi}$$

where $\xi = 0.084 \text{ nm}$, $A = 1600 \text{ eV}$ and $B = 40 \text{ eV}$.

Using the above Morse potential, find the equilibrium bond distance a and the total bonding energy between two lead atoms. Expand the potential around its minimum to determine the Hooke's law spring constant κ associated with the bond. [5]

(c) Consider now a one-dimensional chain of lead atoms held together by springs with spring constant κ and equilibrium spring length a . You should assume the masses move only in one dimension (along the direction of the chain). The atomic weight of a lead atom is 207.0 g mol^{-1} . Determine the sound velocity of this chain. [5]

(d) Expand the potential one order further around the minimum. Explaining your reasoning, using dimensional analysis or otherwise, estimate the thermal expansion coefficient

$$\alpha = \frac{1}{L} \frac{\partial L}{\partial T}$$

where L is the length of a single lead-lead bond. You do not need to compute any dimensionless numerical factors of order unity. You should assume that the temperature is high enough that you can neglect the effects of quantum mechanics, but also the temperature is low enough that you do not need further orders of the expansion around the minimum. [5]

(e) Explaining your reasoning, in what range of temperatures

(i) is it acceptable to neglect higher orders of the expansion around the minimum?

(ii) are the effects of quantum mechanics important? [8]

4. (a) Define the terms *paramagnet*, *diamagnet*, and *ferromagnet*. Give an example of each of these. [3]

(b) The gadolinium-III ion (Gd^{3+}) has 7 electrons in a partially filled f shell. Using Hund's rules, determine the spin S of a Gd^{3+} ion. [2]

Gadolinium nitride (GdN) has a face-centred cubic lattice and its basis has a Gd^{3+} ion at position $[0, 0, 0]$ and an N^{3-} ion at position $[\frac{1}{2}, 0, 0]$ in units of the lattice constant $a = 0.50 \text{ nm}$. Note that N^{3-} is a filled shell configuration.

In a crystal environment it is often the case that spins cannot point in arbitrary directions. For the rest of this problem you should assume that the Gd^{3+} spin can only point in two possible directions, $+\hat{z}$ or $-\hat{z}$.

(c) At high temperature, a very small magnetic field H_z is applied to a sample of GdN in the $\hat{z}$ direction resulting in a magnetization $M_z = \chi H_z$. The susceptibility χ has the form

$$\chi = \frac{C}{T}$$

Determine the value of the coefficient C in SI units. [8]

(d) At temperatures below $T_c = 70 \text{ K}$, the sample of GdN is found to be ferromagnetic. How might the above equation for χ be modified to improve its accuracy for $T > T_c$? At absolute zero temperature, what would you expect the magnetization of the material to be? [5]

(e) Give a sketch of the magnetization in the absence of an external magnetic field as a function of temperature for $0 \leq T < 2T_c$.

A fixed small field, $H_z > 0$, is applied to the system. On the same plot as above, sketch the magnetization as a function of temperature in the presence of this applied field, noting the differences from the $H_z = 0$ case. [7]